EduCred System Explained

Plain-English technical summary for business pitch

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One-line pitch:
EduCred stores normal app data in PostgreSQL, certificate files on IPFS, and tamper-proof certificate hashes on Ethereum Sepolia so anyone can verify if a certificate is real.

1. Big Picture

EduCred is a digital certificate issuing and verification system. A university can issue credentials, students can view them, and employers or verifiers can check authenticity online.

- Frontend: React app in the client folder.
- Backend: Node.js and Express API in the server folder.
- Database: PostgreSQL, hosted through Render.
- Blockchain: Ethereum Sepolia testnet.
- File storage: IPFS through Pinata.

2. What Gets Stored Where

- PostgreSQL stores users, students, universities, certificates, ledger events, audit logs, verification logs, and fraud alerts.
- IPFS stores certificate PDFs and metadata JSON files. Pinata keeps these IPFS files pinned and available.
- Ethereum Sepolia stores only the certificate hash, issuer wallet, timestamp, certificate type, and revoked status.

Simple memory trick:

PostgreSQL = business records and search

IPFS = files and metadata

Blockchain = public proof of authenticity

3. Why Blockchain Is Used

The blockchain is used as a tamper-proof public ledger. EduCred does not put the full certificate or student data on-chain. It puts only a hash, which is like a digital fingerprint.

- If the certificate changes, the hash changes.
- If the hash exists on-chain, the certificate was officially anchored.
- If the hash does not match, the certificate may be fake or modified.

4. Why Ethereum Sepolia Is Used

Sepolia is Ethereum’s test network. It behaves like Ethereum but uses test ETH, so it is suitable for demos, pilots, and pitch-stage validation without real transaction costs.

Pitch wording: EduCred currently uses Ethereum Sepolia as a public test ledger. The same architecture can later move to Ethereum mainnet, Polygon, or a private institutional chain.

5. Smart Contract Role

The smart contract is the on-chain registry. It decides which university wallets are allowed to issue, stores certificate hashes, verifies hashes, and records revocation status.

- addIssuer: authorizes a university wallet.
- storeCertificate: saves the certificate hash on-chain.
- verifyHashFull: checks if a certificate hash exists, who issued it, and when.
- revokeHash: marks a certificate as revoked.

6. Algorithms and Security

- SHA-256 hashing creates certificate fingerprints. Same input gives same hash; changed input gives a totally different hash.
- bcrypt stores passwords safely as password hashes.
- JWT handles login sessions.
- AES-256-GCM encrypts university private keys before storing them.
- UUID and random bytes generate unique certificate IDs.
- Ethereum wallet signatures prove which university issued a certificate.

7. Certificate Issue Flow

```
University submits certificate details
Backend validates data
Backend creates SHA-256 hash
Certificate record is saved in PostgreSQL as pending
Approved certificate hash is sent to Sepolia smart contract
Blockchain returns transaction hash
PostgreSQL record becomes confirmed
PDF is generated and uploaded to IPFS
PDF URL and file hash are stored in PostgreSQL
```

8. Verification Flow

A verifier can verify by certificate ID or by uploading the original PDF.

• By ID: backend finds the certificate in PostgreSQL, gets the hash, and checks it on blockchain.
• By PDF: backend hashes the uploaded PDF, matches it with stored PDF hash, then checks blockchain.
• If valid: system returns certificate metadata and blockchain proof.
• If fake/tampered: system records verification failure and may create a fraud alert.
• If revoked: system clearly says the certificate is revoked.

9. Important Database Tables

• User: login identity, email, role, password hash, OTP fields.
• Student: student profile linked to a user.
• University: institution profile, approval status, wallet address, encrypted private key.
• Certificate: certificate ID, student, course, hash, blockchain transaction hash, IPFS URL, status.
• Ledger: issue/revoke/verify event history.
• AuditLog: security and admin activity.
• VerificationLog: every verification attempt.
• FraudAlert: suspicious certificate attempts.

10. Business Pitch Version

EduCred solves certificate fraud by making every issued credential independently verifiable. The app stores operational records in PostgreSQL, stores certificate files on IPFS, and anchors cryptographic fingerprints on Ethereum Sepolia. This means a verifier does not need to trust a screenshot, PDF, or manual university response. They can check the certificate against a public tamper-proof ledger.

11. Current Production Note

Registration email delivery previously failed because Render free web services block outbound SMTP ports. The code has been updated to support HTTPS email delivery using Resend, while keeping SMTP as a fallback for local or paid environments.

```
Render environment variables needed:
EMAIL_PROVIDER=resend
RESEND_API_KEY=your_key
EMAIL_FROM=EduCred <verified@yourdomain.com>
```